

Francisco Pinto Santos

Software Engineer | Backend & Distributed Systems | PhD

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LinkedIn | GitHub



About Me

Software Engineer with more than seven years of experience in backend development, distributed systems and software architecture design. Throughout my career, I have worked across APIs, Integration, Identity, Big Data, Platform and Observability, combining software development with the evolution, integration and operation of complex platforms. My academic background includes a BSc, two MScs and a PhD in Computer Science and Artificial Intelligence, as well as research and university teaching experience.

Skills

Backend: Java, Spring Boot, Node.js, Express.js, TypeScript, Python, Flask, Django, FastAPI and REST APIs.

Big Data: Apache Kafka, Apache Spark, SQL, PostgreSQL, MySQL, MongoDB, Elasticsearch and Neo4j.

Platform: Bash, Git, GitHub Actions, CI/CD, Docker, Kubernetes, AKS, Helm, Helmfile, AWS and Azure.

Identity: OAuth 2.0, OpenID Connect, CIBA, JWT, JWS, JWE, JWK/JWKS, mTLS and Keycloak.

Observability: OpenTelemetry, Grafana, Loki, Mimir, Tempo and Prometheus.

Architecture and ecosystem: microservices, distributed systems, data-driven architectures, Open Gateway and CAMARA.

Languages

Spanish: native.

English: full professional proficiency.

Experience

Software Engineer - Backend & Distributed Systems - Telefónica Innovación Digital

01/2026 - present · Madrid, Spain

- Design and implementation of cross-functional initiatives on a distributed platform for telco services and data, working across multiple components and coordinating changes with backend, identity, privacy, API, QA and infrastructure teams.
- Implementation of new authentication flows requiring coordinated changes across backend services, administrative APIs, SDKs, configuration, testing and documentation.
- Migration and coexistence of two distributed platforms, ensuring functional compatibility, porting fixes and validating end-to-end flows during the transition to single-tenant and multibrand models.
- Development and evolution of the observability infrastructure for a cloud-native AI computing platform using Kubernetes/AKS, Helm/Helmfile, OpenTelemetry, Grafana, Loki, Mimir and Tempo.
- Development of infrastructure and internal tooling, including the deployment of a technical portal on Kubernetes, automation of documentation and API synchronisation, and the creation of CI/CD pipelines.

Partner Engineer - Telefónica Innovación Digital

01/2024 - 01/2026 · Madrid, Spain

- Design, implementation and validation of REST API integrations and OAuth flows for partners within Open Gateway and CAMARA.

- Development and maintenance of mocks and tooling with Node.js to facilitate testing and integrations.
- Operation of integration environments deployed with Kubernetes on AWS and Azure, including provisioning, configuration and monitoring.
- Diagnosis and resolution of issues in backend microservices and Big Data systems using Apache Spark and Apache Kafka.
- Development of tooling, test collections and technical documentation to facilitate integrations, automate tasks and improve team operations.

Support Engineer at Telefónica - BCNC Group

03/2023 - 01/2024 · Madrid, Spain

- Testing and validation of REST API integrations and OAuth flows for third parties within Open Gateway and CAMARA.
- Development of mocks with Node.js to facilitate integration and testing.

Backend & Architecture R&D Lead - BISITE Research Group

09/2020 - 03/2023 · Salamanca, Spain

- Coordination and technical planning of R&D projects.
- Design and implementation of AWS-based architectures.
- Technical leadership of Java Spring, Node.js and Python microservices, including ERP systems, scrapers, crawlers and AI-model APIs.
- Technical leadership of Apache Kafka-based ETL systems and development of internal tooling.

Full-stack Developer - BISITE Research Group

06/2019 - 09/2020 · Salamanca, Spain

- Design and implementation of Java Spring, Node.js and Python microservices, including ERP systems, scrapers, crawlers and AI-model APIs.
- Design and development of Apache Kafka-based ETL systems and web applications using the MEVN stack.
- Development of internal tooling for code generation and CI automation with GitHub Actions.
- Generation of datasets through web scraping and crawling, and development of Python packages, including deepint.

Intern - BISITE Research Group

06/2017 - 06/2019 · Salamanca, Spain

- Development of REST APIs with Java Spring.
- Development of web applications using the MEVN stack.

Lecturer - University of Salamanca

01/2021 - present · Salamanca, Spain

- Lecturer on the Master's Degree in Generative AI Project Management, 2024/2025 edition, teaching Generative AI Project Design.
- Lecturer on the Master's Degree in Smart Cities & Intelligent Buildings, 2023/2024 edition, teaching Smart Cities Asset Management.
- Lecturer in the 2022 edition, teaching introductory artificial intelligence, convolutional neural networks and recurrent neural networks, in collaboration with the Sociedad de Promoción Económica de Gran Canaria (SPEGC).
- Lecturer in the 2021-2024 editions of the Master's Degree in Internet of Things, covering IoT architectures and introductory artificial intelligence.

Lecturer - Pontifical University of Salamanca

02/2022 - 11/2022 · Salamanca, Spain

- Teaching in Databases management and Fundamentals of Computer Science I and II.

Education

PhD in Computer Science and Artificial Intelligence - University of Salamanca

10/2022 - 07/2026 · Salamanca, Spain

- Grade: Summa cum laude.
- Thesis: *Trustworthy Distributed Architectures for Intelligent Information Processing*.
- Research focused on distributed architectures for real-time intelligent processing, with scalability, software security and privacy by design.

MSc in Smart Cities & Intelligent Buildings - University of Salamanca

09/2021 - 09/2022 · Salamanca, Spain

- GPA: 9.04/10.

MSc in Computer Engineering - University of Salamanca

09/2020 - 09/2022 · Salamanca, Spain

- GPA: 9.57/10.
- First in the 2020-2022 class.
- Best academic record award for the master's degree and award from the Faculty of Sciences.

BSc in Computer Engineering - University of Salamanca

09/2016 - 07/2020 · Salamanca, Spain

- GPA: 9.55/10.
- First in the 2016-2020 class.
- Best academic record award and award from the Faculty of Sciences.
- Third place in SEDEA's national ranking of engineering students.
- Student representative for three years.

Achievements

- **04/2026:** third place in the University of Salamanca Alumni Talent Award.
- **01/2023:** best academic record award for the 2020-2022 Master's Degree in Computer Engineering class at the University of Salamanca.
- **11/2022:** award for one of the best academic records in the 2020-2022 Faculty of Sciences class at the University of Salamanca.
- **04/2022:** individual mention in the Microsoft Azure hackathon organised by Viewnext IBM.
- **11/2021:** award for one of the best academic records in the 2016-2020 Faculty of Sciences class at the University of Salamanca.
- **07/2021:** best academic record award for the 2016-2020 BSc in Computer Engineering class at the University of Salamanca.
- **07/2020:** third place in SEDEA's national ranking of engineering students.
- **02/2020:** first Google Hash Code team at Everis Salamanca Hub.
- **10/2019:** winner of the Blockchain hackathon organised by the DIGITEC project.
- **10/2019:** Global Nominee at the NASA Space Apps Challenge, Everis Salamanca Hub.
- **02/2019:** winner of the Smart Speaker Hackathon organised by Viewnext.